

Monetary Policy Transmission

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1. Introduction & Data

In this project, I study the impact of monetary policy shocks on inflation and output in the US economy. The macroeconomic data was retrieved from the FRED database in `scripts/01_data_prep_US.R`.

```
macro_data <- read_rds("data/cleaned_macro_data_US.rds")

var_data <- macro_data %>%
  dplyr::select(comm_growth, output_growth, infl, rate_level, reserve_growth, m1_growth)
```

For the regression we take data of :

- Commodity price growth
- Industrial output growth
- Inflation rate
- Federal Funds Rate (policy instrument)
- Non-borrowed reserves growth
- M1 money supply growth

2. VAR and LP models

```
lag_selection <- VARselect(var_data, lag.max = 12, type = "const")

print(lag_selection$selection)

## AIC(n)  HQ(n)  SC(n) FPE(n)
##      3      1      1      3

p_lag <- lag_selection$selection["SC(n)"]

var_model <- vars::VAR(var_data, p = p_lag, type = "const")

roots <- roots(var_model)
print("Number of roots greater than 1:")

## [1] "Number of roots greater than 1:"
print(sum(roots>=1))

## [1] 0
```

Schwartz information criteria suggest lag 1 for our model. All roots of characteristic equation is smaller than 1, therefore, data is stationary

```
irf_result <- irf(var_model,
  impulse = "rate_level",
  response = c("output_growth", "infl"),
  n.ahead = 60,
  boot = TRUE,
  ci = 0.95,
  runs = 100)

extract_irf_to_df <- function(irf_obj, shock_name) {

  responses <- colnames(irf_obj$irf[[shock_name]])

  df_list <- lapply(responses, function(resp) {
    data.frame(
      horizon = 0:(nrow(irf_obj$irf[[shock_name]]) - 1),
      response_var = resp,
      pe = irf_obj$irf[[shock_name]][, resp],      # estimation
      lower = irf_obj$Lower[[shock_name]][, resp], # lower bound of CI
      upper = irf_obj$Upper[[shock_name]][, resp]  # upper bound of CI
    )
  })

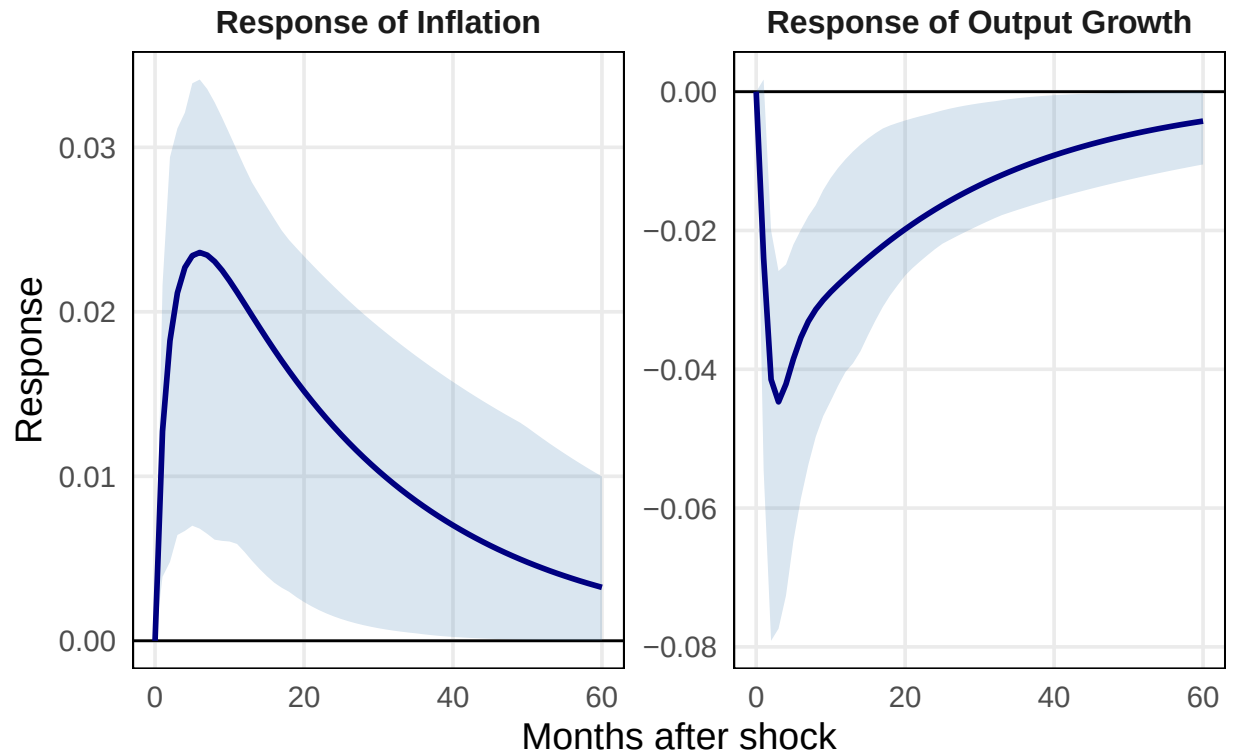
  bind_rows(df_list) %>% as_tibble()
}

irf_data <- extract_irf_to_df(irf_result, "rate_level")
```

Calculate impulse response function for 60 lags, using 95% confidence interval and bootstrap via 1000 runs.

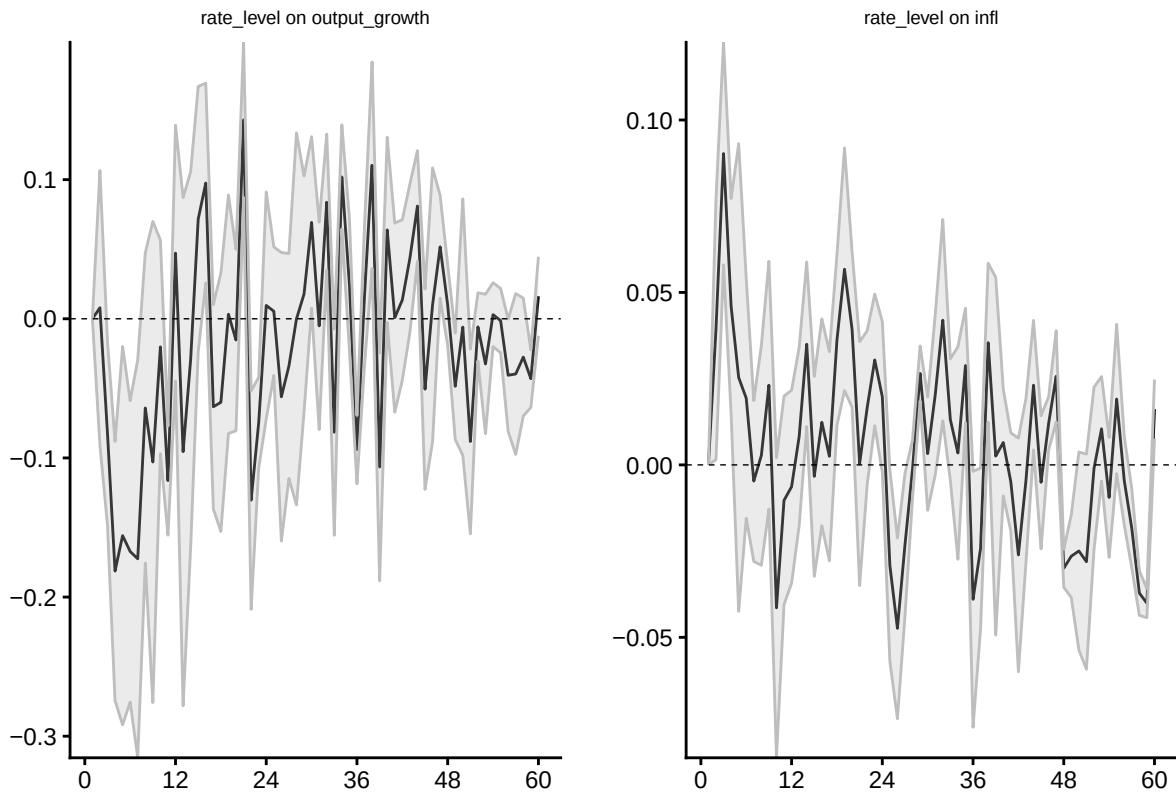
```
ggplot(irf_data, aes(x = horizon, y = pe)) +
  geom_hline(yintercept = 0, color = "black", linetype = "solid", linewidth = 0.5) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2, fill = "steelblue") +
  geom_line(color = "navy", linewidth = 1) +
  facet_wrap(~ response_var, scales = "free_y",
    labeller = as_labeller(c(
      output_growth = "Response of Output Growth",
      infl = "Response of Inflation"
    ))) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Impulse Responses to an Interest Rate Shock",
    x = "Months after shock",
    y = "Response"
  ) +
  theme(
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.5)
  )
```

Impulse Responses to an Interest Rate Shock



As we can observe, monetary policy shock gives immediate effect on output, while effect on inflation is statistically insignificant, and even increase right after shock.

```
lp_results <- lp_lin(  
  endog_data = var_data,  
  lags_endog_lin = 4,  
  trend = 0,  
  shock_type = 1,  
  confint = 1.96,  
  hor = 60  
)  
  
plot_lin_graphs <- plot_lin(lp_results)  
  
plot_output <- plot_lin_graphs[[10]]  
plot_infl <- plot_lin_graphs[[16]]  
  
plot_output + plot_infl & scale_x_continuous(breaks = seq(0, 60, by = 12))
```

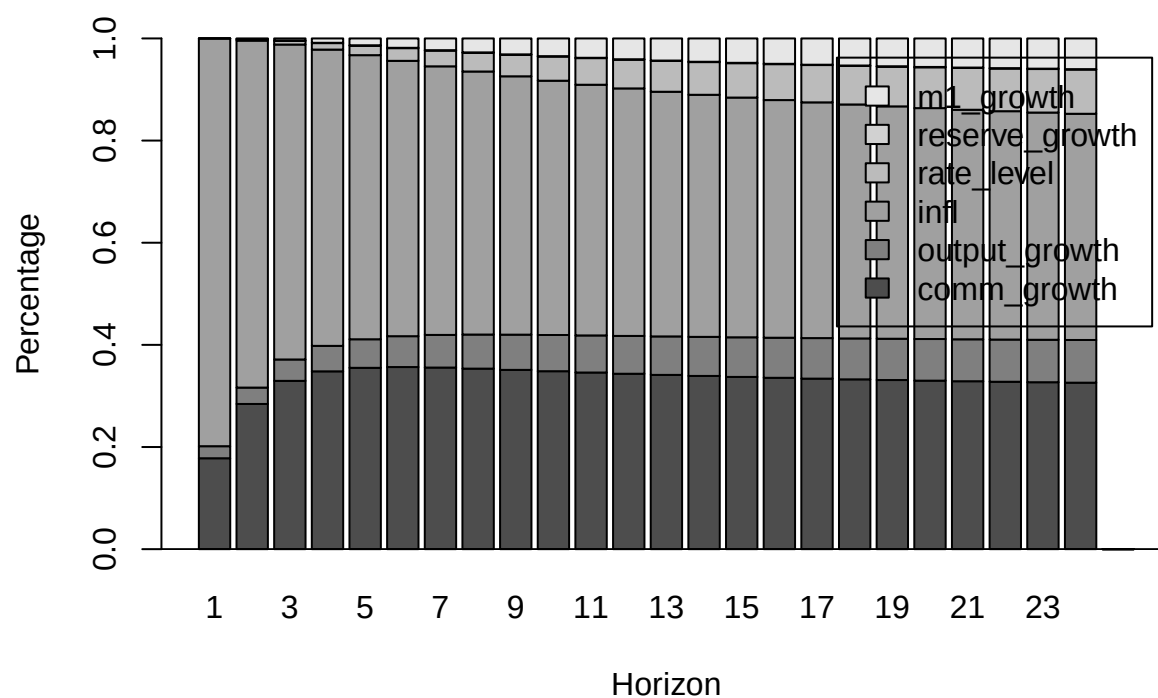


Use Local Projections method for identification. Still impact of rate shock on inflation is negligible, or spikes up followed by spikes down like a chain with an overall slightly decreasing trend.

```
fevd_results <- fevd(var_model, n.ahead = 24)
```

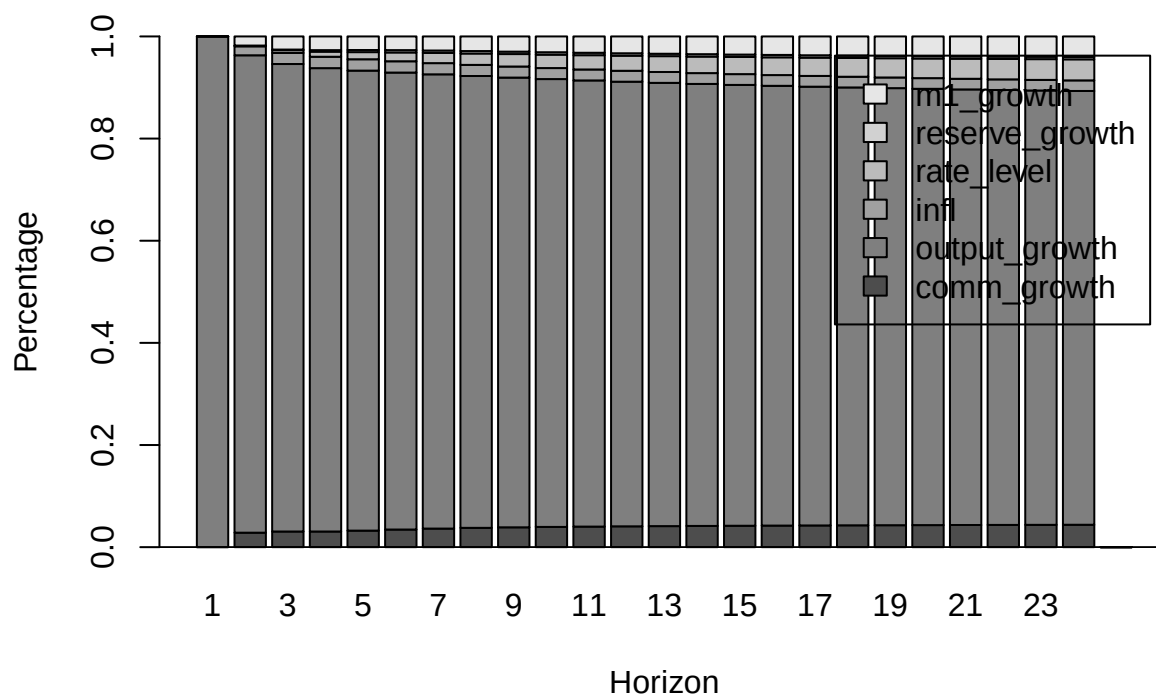
```
plot(fevd_results, names = "infl", main = "Variance Decomposition of Inflation")
```

Variance Decomposition of Inflation



```
plot(fevd_results, names = "output_growth", main = "Variance Decomposition of Output")
```

Variance Decomposition of Output



Overall, rate change contributes to 5-7% of variance of inflation and output change.

```
causality_infl <- causality(var_model, cause = "rate_level")$Granger
print(causality_infl)
```

```
##
## Granger causality H0: rate_level do not Granger-cause comm_growth
## output_growth infl reserve_growth m1_growth
##
## data: VAR object var_model
## F-Test = 5.423, df1 = 5, df2 = 1170, p-value = 6.129e-05
```

Testing causality - we reject the null hypotheses that rate do not Granger-cause

```
amat <- diag(6)
amat[lower.tri(amat)] <- NA

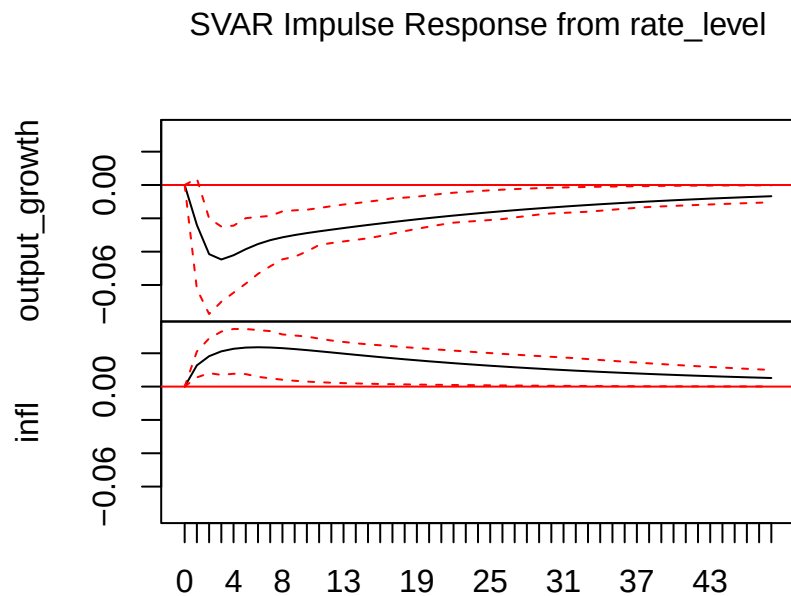
bmat <- diag(6)
diag(bmat) <- NA

svar_model <- vars::SVAR(var_model,
  estmethod = "scoring",
  Amat = amat,
  Bmat = bmat)

irf_svar <- irf(svar_model,
  impulse = "rate_level",
```

```
response = c("output_growth", "infl"),
n.ahead = 48,
boot = TRUE,
runs = 100,
ci = 0.9)
```

```
plot(irf_svar)
```



90 % Bootstrap CI, 100 runs

When use sVAR for identification of monetary policy shock, we can notice that, while there is still significant inflation increase in the beginning, after 18th month inflation slowly goes down, however still insignificantly.

3. Romer's residuals

```
romer_monthly <- read_delim("RomerandRomerDataAppendix.csv", locale = locale(decimal_mark = ","))

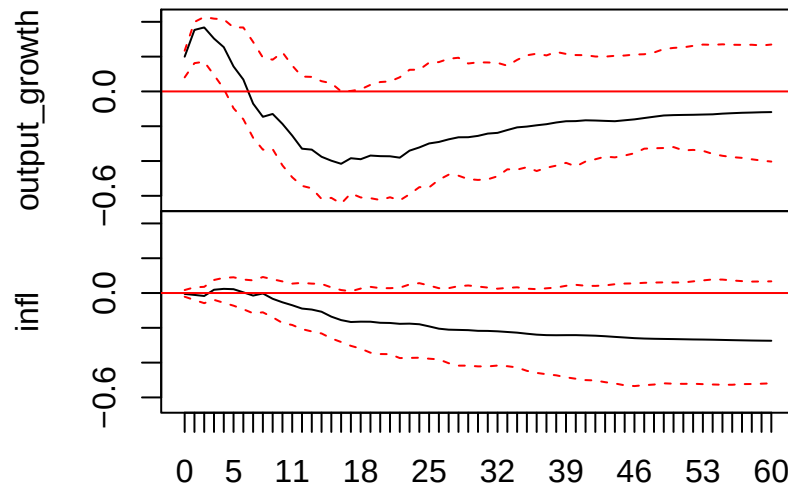
romer_clean <- romer_monthly %>%
  mutate(
    time = dmy(DATE)
  ) %>%
  dplyr::select(time, romer_inv = RESID) %>%
  arrange(time)
```

For the further identification of monetary policy shock, we will use Romer's residuals - change between real and predicted rates.

Now, instead of percent change, we will observe price level reaction to policy shock.

```
var_romer_data <- final_us_data %>%  
  dplyr::select(romer_inv, comm_growth, output_growth, infl, reserve_growth, m1_growth)  
  
var_romer_data <- as.data.frame(var_romer_data)  
  
var_romer_model <- vars::VAR(var_romer_data, p = 12, type = "const")  
  
irf_romer_cum <- irf(var_romer_model,  
  impulse = "romer_inv",  
  response = c("output_growth", "infl"),  
  n.ahead = 60,  
  cumulative = TRUE,  
  boot = TRUE,  
  ci = 0.9,  
  runs = 100)  
  
plot(irf_romer_cum)
```

Orthogonal Impulse Response from romer_inv (cumulative)



90 % Bootstrap CI, 100 runs

Price level falls after 8th month, however effect is insignificant, as well as for output.

```
amat <- diag(6)  
amat[lower.tri(amat)] <- NA  
  
bmat <- diag(6)  
diag(bmat) <- NA
```



```

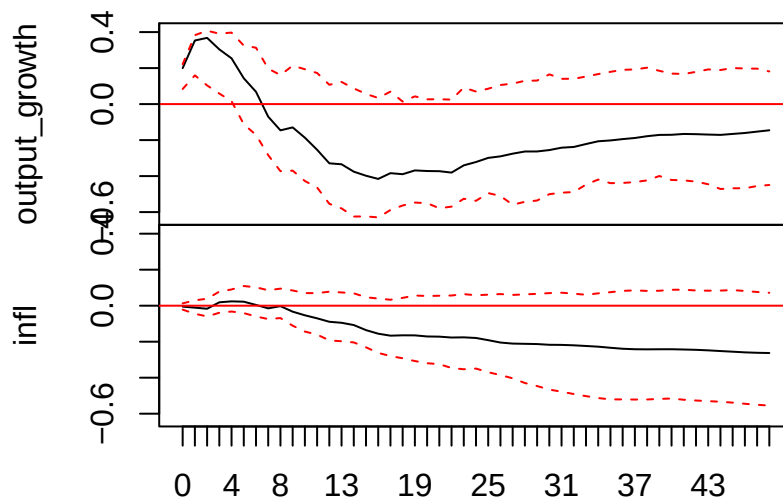
svar_romer_model <- vars::SVAR(var_romer_model,
                               estmethod = "scoring",
                               Amat = amat,
                               Bmat = bmat)

irf_svar_romer <- irf(svar_romer_model,
                      impulse = "romer_inv",
                      response = c("output_growth", "infl"),
                      n.ahead = 48,
                      cumulative = TRUE,
                      boot = TRUE,
                      ci = 0.9,
                      runs = 100)

plot(irf_svar_romer)

```

SVAR Impulse Response from romer_inv (cumulative)



90 % Bootstrap CI, 100 runs

Now, we see statistically significant drop for output and decrease for price level. While upper confidence interval bound for prices is slightly above zero, it may be due to strict z-critical as argue Sims&Zha(1999) - more important is the fact that there is no raise of inflation.

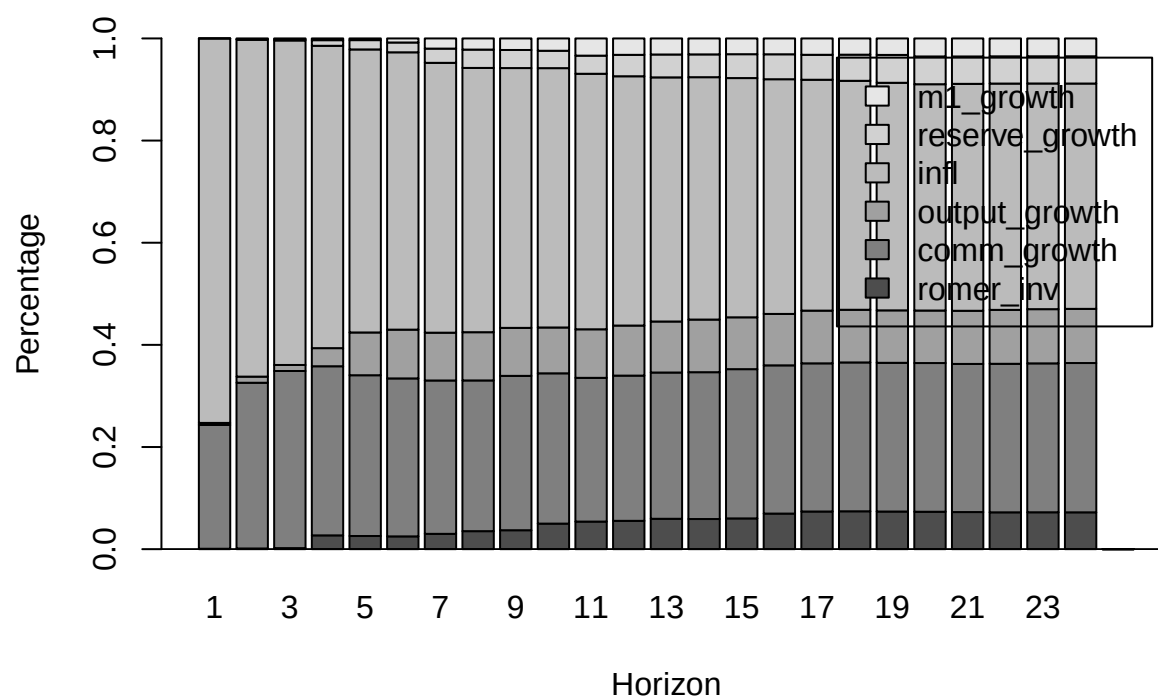
```

fevd_romer_results <- fevd(var_romer_model, n.ahead = 24)

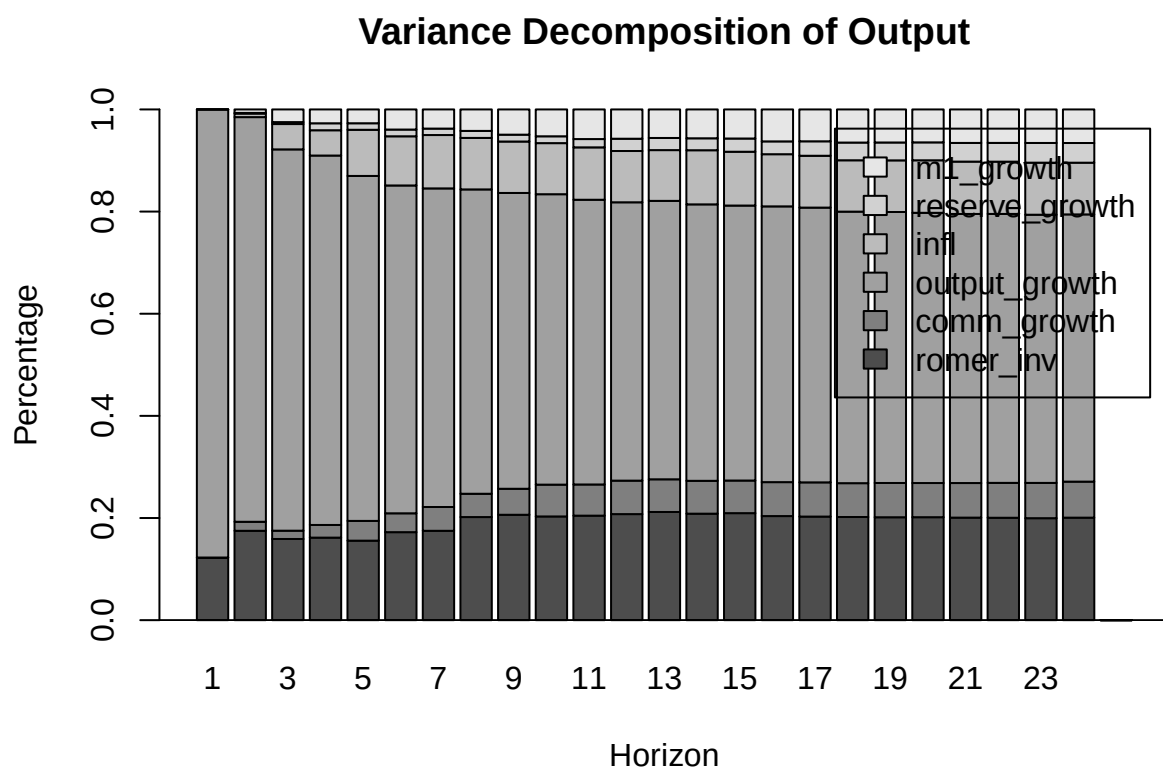
plot(fevd_romer_results, names = "infl", main = "Variance Decomposition of Inflation")

```

Variance Decomposition of Inflation



```
plot(fevd_romer_results, names = "output_growth", main = "Variance Decomposition of Output")
```



For inflation, this setting after 4th lag explain about 6-7% of variance, while for output this value is about 20%

Literature

- **Federal Reserve Bank of St. Louis.** *Federal Reserve Economic Data (FRED)*. URL: <https://fred.stlouisfed.org/>
- **Romer, C. D., & Romer, D. H. (2004).** A New Measure of Monetary Shocks: Derivation and Implications. *The American Economic Review*, 94(4), 1055-1084. URL: <https://www.aeaweb.org/articles?id=10.1257/0002828042002651>
- **Sims, C. A., & Zha, T. (1999).** Error Bands for Impulse Responses. *Econometrica*, 67(5), 1113-1155.